



TEMPEST 2025 AND PRAYAAN 6.0

HIGH-SPEED PLANING HULL INTEGRATED WITH AN AZIMUTHAL PROPULSION SYSTEM FOR SUPERIOR MANEUVERABILITY. IT'S CONTROLLED VIA A 6-CHANNEL RC SYSTEM WITH PID TUNING, ENABLING PRECISION THRUST VECTORING AND AGILE NAVIGATION



IN SEARCH OF INNOVATION

BUILDNEO: A student-led research and development initiative at the Indian Maritime University, Visakhapatnam, focused on innovation and practical skill development in naval architecture. Founded by ambitious alumni, it thrives on a network of past and present students collaborating on working prototypes and research. The initiative features a dynamic leadership team, including roles in external communications, electronics, technical documentation, coding, CAD modeling, fabrication, additive manufacturing, and social media. By promoting project flexibility and creativity, BuildNEO prepares members for academic and industrial challenges, aiming to become a hub of talent and innovation in the maritime industry.

WHITE 3.4

Our project involves the development of a compact, remotely operated surface vehicle designed for short-range aquatic missions. The system is powered by an efficient brushless DC motor and is controlled wirelessly using a reliable radio frequency module. The vehicle is built to navigate in water with precision and endurance, making it suitable for tasks such as environmental monitoring, aquatic research, or surface-level inspections. The streamlined design ensures stable performance, and the integration of high-efficiency components enables effective power management and operational reliability. This project emphasizes modularity, ease of control, and adaptability for various real-world applications in controlled water environments.

OBJECTIVE

To design and develop a small, efficient, and remotely operated watercraft capable of performing short-range surface-level tasks, with reliable control and stable performance in calm water conditions.

The project aims to demonstrate effective integration of propulsion, power, and control systems in a compact form.

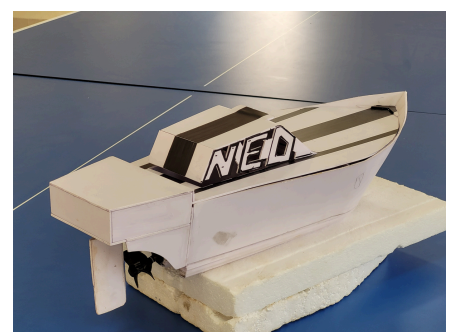
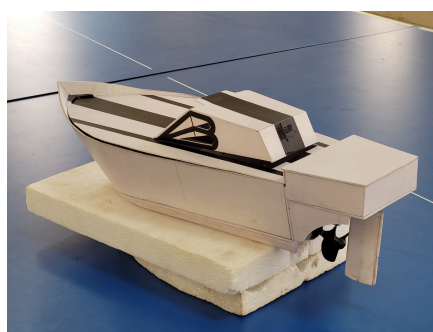
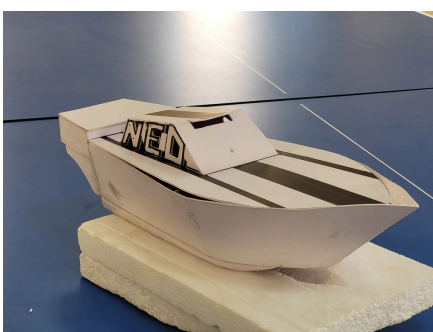
It also serves as a platform for future enhancements such as automation, data collection, or sensor-based navigation.

SPECIFICATIONS

- 55 cm (L) × 21 cm (B) × 5 cm (Draft)
- A2212 1000KV BLDC motor
- SimonK 30A ESC
- 11.1V 2200mAh 35C LiPo battery
- FlySky FS-iA6B 2.4GHz 6CH receiver
- Max speed: 3.4 knots
- Range: 600 m
- Endurance: 30–70 minutes

FEATURES

The White 3.4 is a compact, remotely operated surface vehicle designed for efficient performance and maneuverability in small water bodies. With a hull measuring 55 cm in length, 21 cm in breadth, and a shallow draft of 5 cm, it is optimized for shallow or constrained environments. Powered by an A2212 1000KV BLDC motor and a SimonK 30A ESC, it utilizes an 11.1V 2200mAh 35C LiPo battery to achieve a maximum speed of 3.4 knots and an endurance of 30 to 70 minutes. The vessel is remotely controlled via a FlySky FS-iA6B 2.4GHz 6-channel receiver, offering a reliable operational range of up to 600 meters. This setup makes the White 3.4 ideal for applications such as water-based monitoring, educational robotics, or as a testing platform for autonomous navigation systems.



WHITE 3.8

White 3.8 is a lightweight, remote-controlled watercraft designed for smooth and stable operation in various aquatic environments. Its compact design allows easy navigation in shallow or restricted waters, making it ideal for exploration, research, and hobby use. The craft is powered by a reliable brushless motor system and offers responsive control with extended operational time. With a long-range wireless receiver, it provides consistent performance for medium-distance missions. White 3.8 is a versatile platform that can be adapted for monitoring, testing, or educational applications.

OBJECTIVE

To develop a compact, remotely operated surface vehicle capable of stable and efficient navigation in shallow water environments, with extended operational time and reliable control range, suitable for applications such as aquatic monitoring, research, testing, and educational purposes. The project aims to integrate lightweight components and a user-friendly control system for ease of deployment and operation. It also serves as a scalable platform for future enhancements, including autonomous navigation and sensor integration.

SPECIFICATIONS

- 49 cm × 17 cm × 6 cm (L × B × Draft)
- A2212 1000KV BLDC motor
- SimonK 30A ESC
- 11.1V 2200mAh 35C LiPo battery
- FlySky FS-IA6B 2.4GHz 6CH receiver
- Max speed: 3.8 knots
- Range: 600 m
- Endurance: 40–80 minutes

FEATURES

The White 3.8 is a highly efficient, remote-controlled surface vehicle designed for stable navigation in shallow and confined water environments. It boasts a compact and lightweight design, making it ideal for applications such as aquatic exploration, research, and monitoring. Powered by a reliable A2212 1000KV brushless motor and equipped with a SimonK 30A ESC, it offers a smooth and responsive performance. The vehicle is controlled via a 2.4GHz 6-channel FlySky FS-IA6B receiver, providing a range of up to 600 meters. With an impressive operational endurance of 40 to 80 minutes, the White 3.8 ensures extended mission times, making it an ideal choice for medium-range tasks. Its versatility also allows for future upgrades, including autonomous features and additional sensor integration.

